

CHAPTER 10

Group Homomorphisms

1. Note that $\det(AB) = (\det A)(\det B)$.
2. Observe that $|ab| = |a||b|$.
3. Note that $(f + g)' = f' + g'$.
4. Let E denote any even permutation and O any odd. Observe that $\phi(E) = 0$ and $\phi(O) = 1$. $\phi(E) = 0 = 0 + 0 = \phi(E) + \phi(E)$. $\phi(O) = 1 = 0 + 1 = \phi(E) + \phi(O)$. The other cases are similar.
5. Observe for every positive integer r we have $(xy)^r = x^r y^r$ so the mapping is a homomorphism. When r is odd the kernel is $\{1\}$ so the mapping is one-to-one and an isomorphism. When n is even the kernel is $\{\pm 1\}$ and the mapping is two-to-one.
6. Recall, $\int(f + g) = \int f + \int g$. Kernel = $\{0\}$. No.
7. $(\sigma\phi)(g_1 g_2) = \sigma(\phi(g_1 g_2)) = \sigma(\phi(g_1)\phi(g_2)) = \sigma(\phi(g_1))\sigma(\phi(g_2)) = (\sigma\phi)(g_1)(\sigma\phi)(g_2)$. It follows from Theorem 10.3 that $|G/\text{Ker } \phi| = |H|$ and $|G/\text{Ker } \sigma\phi| = |K|$. Thus, $[\text{Ker } \sigma\phi : \text{Ker } \phi] = |\text{Ker } \sigma\phi / \text{Ker } \phi| = |H|/|K|$.
8. See Exercise 20 of Chapter 5. The kernel is the set of even permutations in G . When G is S_n the kernel is A_n and from Theorem 10.3 we have that S_n/A_n is isomorphic to $\{+1, -1\}$. So, A_n has index 2 in S_n and is normal in S_n . The kernel is the subgroup of even permutations in G . If the members of G are not all even then the coset other than the kernel is the set of odd permutations in G . All cosets have the same size.
9. $\phi((g, h)(g', h')) = \phi((gg', hh')) = gg' = \phi((g, h))\phi((g', h'))$. The kernel is $\{(e, h) \mid h \in H\}$.
10. See Exercise 9 of Chapter 1. The kernel is the subgroup of rotations in G . If the members of G are not all rotations then the coset other than the kernel is the set of reflections in G . All cosets have the same size.
11. The mapping $\phi : Z \oplus Z \rightarrow Z_a \oplus Z_b$ given by $\phi((x, y)) = (x \bmod a, y \bmod b)$ is operation preserving by Exercise 9 in Chapter 0. If $(x, y) \in \text{Ker } \phi$, then $x \in \langle a \rangle$ and $y \in \langle b \rangle$. So, $(x, y) \in \langle (a, 0) \rangle \times \langle (0, b) \rangle$. Conversely, every element in $\langle (a, 0) \rangle \times \langle (0, b) \rangle$ is in $\text{Ker } \phi$. So, by Theorem 10.3, $Z \oplus Z \rightarrow Z_a \oplus Z_b$ is isomorphic to $\langle (a, 0) \rangle \times \langle (0, b) \rangle$.
12. $x \rightarrow x \bmod k$ is a homomorphism with kernel $\langle k \rangle$.

13. $(a, b) \rightarrow b$ is a homomorphism from $A \oplus B$ onto B with kernel $A \oplus \{e\}$. So, by Theorem 10.3, $(A \oplus B)/(A \oplus \{e\}) \approx B$. Chapter 5. The kernel is the set of even permutations in G . When G is S_n the kernel is A_n and from Theorem 10.3 we have that S_n/A_n is isomorphic to $\{+1, -1\}$. So, A_n has index 2 in S_n and is normal in S_n . The kernel is the subgroup of even permutations in G . If the members of G are not all even then the coset other than the kernel is the set of odd permutations in G . All cosets have the same size.
14. Observe that $\phi(6 + 7) = \phi(1) = 3$ while $\phi(6) + \phi(7) = 8 + 1 = 9$.
15. By property 6 of Theorem 10.1, we know $\phi^{-1}(9) = 23 + \text{Ker } \phi = \{23, 3, 13\}$.
16. Observe that such a mapping would be an isomorphism and isomorphisms preserve order.
17. Suppose ϕ is such a homomorphism. By Theorem 10.3, $\text{Ker } \phi = \langle(8, 1)\rangle, \langle(0, 1)\rangle$ or $\langle(0, 1)\rangle$. In these cases, the element $(1, 0) + \text{Ker } \phi$ in $(Z_{16} \oplus Z_2)/\text{Ker } \phi$ has order either 16 or 8. So, $(Z_{16} \oplus Z_2)/\text{Ker } \phi$ is not isomorphic to $Z_4 \oplus Z_4$.
18. No, because of part 3 of Theorem 10.1. No, because the homomorphic image of a cycle group must be cyclic.
19. Since $|\text{Ker } \phi|$ is not 1 and divides 17, ϕ is the trivial map.
20. 0 onto Z_8 ; 4 to Z_8 .
21. By Theorem 10.3 we know that $|Z_{30}/\text{Ker } \phi| = 5$. So, $|\text{Ker } \phi| = 6$. The only subgroup of Z_{30} of order 6 is $\langle 5 \rangle$.
22. Let $|\phi(g)| = 8$. By Theorem 10.1 part 3, $|g| = 8k$. Then $|g^k| = 8$. To generalize replace 8 by n .
23. $|\phi^{-1}(H)| = |H||\text{Ker } \phi|$.
24.
 - a. Let $\phi(1) = k$. Then $\phi(7) = 7k \bmod 15 = 6$ so that $k = 3$ and $\phi(x) = 3x$.
 - b. $\langle 3 \rangle$.
 - c. $\langle 5 \rangle$.
 - d. $1 + \langle 5 \rangle$.
25. To define a homomorphism from Z_{20} onto Z_{10} we must map 1 to a generator of Z_{10} . Since there are four generators of Z_{10} we have four homomorphisms. (Once we specify that 1 maps to an element a , the homomorphism is $x \rightarrow xa$.) To define a homomorphism from Z_{20} to Z_{10} we can map 1 to any element of Z_{10} . (Be careful here, these mappings are well defined only because 10 divides 20.)

26. There are four: $x \rightarrow (x \bmod 2, 0)$; $x \rightarrow (0, x \bmod 2)$;
 $x \rightarrow (x \bmod 2, x \bmod 2)$; $x \rightarrow (0, 0)$.
27. If ϕ is a homomorphism from Z_n to Z_n with $\phi(1) = k$, then by property 2 of Theorem 10.1 $\phi(x) = kx$. Moreover, for each k with $0 \leq k \leq n-1$, the mapping $\phi(x) = kx$ is a homomorphism.
28. $\text{Ker } \phi = A_4$. The trivial homomorphism and the one given in Example 11 are the only homomorphisms. To verify this use Theorem 10.3 and Exercise 70 of Chapter 9.
29. Say the kernel of the homomorphism is K . By Theorem 10.3, $G/K \approx Z_{10}$. So, $|G| = 10|K|$. In Z_{10} , let $\bar{H} = \langle 2 \rangle$. By properties 5, 7, and 8 of Theorem 10.2, $\phi^{-1}(\bar{H})$ is a normal subgroup of G of order $2|K|$. So, $\phi^{-1}(\bar{H})$ has index 2. To show that there is a subgroup of G of index 5, use the same argument with $\bar{H} = \langle 5 \rangle$. If there is a homomorphism from a finite group G onto Z_n , then the same argument shows that G has a normal subgroup of index d for any divisor D of n .
30. Use parts 5 and 8 of Theorem 10.2.
31. By property 6 of Theorem 10.1, $\phi^{-1}(7) = 7\text{Ker } \phi = \{7, 17\}$.
32. Write $U(30) = U_3(30) \times U_{10}(30)$. Then from Exercise 9 we have that $\phi(ab) = a$ is a homomorphism with $\text{Ker } \phi = \{1, 11\}$ and $\phi(7) = 7$.
33. By property 6 of Theorem 10.1, $\phi^{-1}(11) = 11\text{Ker } \phi = \{11, 19, 27, 3\}$.
34. If ϕ were such a homomorphism then by Theorem 10.3 $|\text{Ker } \phi| = 6$ whereas A_4 has no subgroup of order 6 (see Example 5 Chapter 7).
35. $\phi((a, b) + (c, d)) = \phi((a + c, b + d)) = (a + c) - (b + d) = (a - b) + (c - d) = \phi((a, b)) + \phi((c, d))$. $\text{Ker } \phi = \{(a, a) \mid a \in Z\}$.
 $\phi^{-1}(3) = \{(a + 3, a) \mid a \in Z\}$.
36. $4a - 4b$.
37. Consider the mapping ϕ from C^* onto R^+ , given by $\phi(x) = |x|$. (Recall from Chapter 0 that $|a + bi| = \sqrt{a^2 + b^2}$.) By straight forward algebra we have $|xy| = |x||y|$. Thus ϕ is a homomorphism with $\text{Ker } \phi = H$. So, by Theorem 10.3, C^*/H is isomorphic to R^+ .
38. $\text{Ker } \gamma = \text{Ker } \alpha \oplus \text{Ker } \beta$.
39. $\phi(xy) = (xy)^6 = x^6y^6 = \phi(x)\phi(y)$. $\text{Ker } \phi = \langle \cos 60^\circ + i \sin 60^\circ \rangle$.
40. $\langle 12 \rangle$; $\langle 12 \rangle$; in general, the kernel is $\langle \text{lcm}(m, n) \rangle$.
41. Consider the mapping ϕ from K to KN/N given by $\phi(k) = kN$. Since $\phi(kk') = kk'N = kNk'N = \phi(k)\phi(k')$ and $kN \in KN/N$, ϕ is a homomorphism. Moreover, $\text{Ker } \phi = K \cap H$. So, by Theorem 10.3, $K/(K \cap N) \approx KN/N$.

42. Show that the mapping from G/N to G/M given by $gN \rightarrow gM$ is an onto homomorphism with kernel M/N .
43. Since the eight elements of A_4 of order 3 must map to an element of order that divides 3, by Lagrange's Theorem, each of them must map to the identity. But then the kernel has at least 8 elements its order and divides 12. So, the kernel has order 12.
44. $U_k(n)$ is the kernel.
45. Let N be a normal subgroup of D_4 . By Lagrange's Theorem the only possibilities for $|N|$ are 1, 2, 4, and 8. By Theorem 10.4, the homomorphic images of D_4 are the same as the factor groups D_4/N of D_4 . When $|N| = 1$, we know $N = \{e\}$ and $D_4/N \approx D_4$. When $|N| = 2$, then $N = \{R_0, R_{180}\}$, since this is the only normal subgroup of D_4 of order 2, and $D_4/N \approx Z_2 \oplus Z_2$ because D_4/N is a group of order 4 with three elements of order 2. When $|N| = 4$, $|D_4/N| = 2$ so $D_4/N \approx Z_2$. When $|N| = 8$, we have $D_4/N \approx \{e\}$.
46. Use Theorem 10.4 and part 3 of Theorem 10.1.
47. It is divisible by 10. In general, if Z_n is the homomorphic image of G , then $|G|$ is divisible by n .
48. It is divisible by 30. In general, the order of G is divisible by the least common multiple of the orders of all its homomorphic images.
49. It is infinite. Z
50. Let A be the coefficient matrix of the system. If A is an $n \times m$ matrix, then matrix multiplication by A is a homomorphism from $\mathbf{R}^m$ into $\mathbf{R}^n$ whose kernel is S .
51. Let γ be a natural homomorphism from G onto G/N . Let $\overline{H}$ be a subgroup of G/N and let $\gamma^{-1}(\overline{H}) = H$. Then H is a subgroup of G and $H/N = \gamma(H) = \gamma(\gamma^{-1}(\overline{H})) = \overline{H}$.
52. Use Theorem 10.1, part 2.
53. The mapping $g \rightarrow \phi_g$ is a homomorphism with kernel $Z(G)$.
54. **a.** Since $4 = |Z_2 \oplus Z_2|$ does not divide $|D_5|$ there are none.
b. There are four. In addition to the trivial homomorphism, we can map all rotations to the identity and all reflections to any one of the three elements of order 2.
55. Since $(f + g)(3) = f(3) + g(3)$, the mapping is a homomorphism. The kernel is the set of elements in $Z[x]$ whose graphs pass through the point $(3, 0)$. 3 can be replaced by any integer.

56. For the first part use trig identities. The kernel is $\langle 2\pi \rangle$.
57. Let g belong to G . Since $\phi(g)$ belongs to $Z_2 \oplus Z_2 = \langle (1, 0) \rangle \cup \langle (0, 1) \rangle \cup \langle (1, 1) \rangle$, it follows that $G = \phi^{-1}(\langle (1, 0) \rangle) \cup \phi^{-1}(\langle (0, 1) \rangle) \cup \phi^{-1}(\langle (1, 1) \rangle)$. Moreover, each of these three subgroups is proper since ϕ is onto and each is normal by property 8 of Theorem 10.2.
58. Try $g \rightarrow (gH, gK)$.
59. Note that if $z \in Z(G)$ then for all $x \in G$, we have $\phi(x)\phi(z) = \phi(xz) = \phi(zx) = \phi(z)\phi(x)$. Since ϕ is onto H , we have $\phi(z) \in Z(H)$.
60. Since $\phi(Z(D_{12})) \subseteq Z(D_3) = \{R_0\}$, we know $\phi(R_{180}) = R_0$.
61. Let G be a group of order 77. By Lagrange's Theorem every nonidentity of G has order 7, 11, or 77. If G has an element of order 77, then G is cyclic. So, we may assume that all nonidentity elements of G have order 7 or 11. Not all nonidentity elements can have order 11 because, by the Corollary of Theorem 4.4, the number of such elements is a multiple of 10. Not all nonidentity elements of G can have order 7 because the number of such elements is a multiple of 6. So, G must have elements a and b such that $|a| = 11$ and $|b| = 7$. Let $H = \langle a \rangle$. Then H is the only subgroup of G of order 11 for if K is another one then by Theorem 7.2 $|HK| = |H||K|/|H \cap K| = 11 \cdot 11/1 = 121$. But HK is a subset of G and G only has 77 elements. Because for every x in G , xHx^{-1} is also a subgroup of G of order 11, we must have $xHx^{-1} = H$. So, $N(H) = G$. Since H has prime order, H is cyclic and therefore Abelian. This implies that $C(H)$ contains H . So, 11 divides $|C(H)|$ and $|C(H)|$ divides 77. This implies that $C(H) = G$ or $C(H) = H$. If $C(H) = G$, then $|ab| = 77$. If $C(H) = H$, then $|N(H)/C(H)| = 7$. But by the "N/C" Theorem (Example 16) $N(H)/C(H)$ is isomorphic to a subgroup of $\text{Aut}(H) \approx \text{Aut}(Z_{11}) \approx U(11)$ (see Theorem 6.5). Since $U(11) = 10$, we have a contradiction.
62. There are no homomorphisms from Z onto S_3 since the image of a cyclic group must be cyclic. For each element x in S_3 the mapping from Z to S_3 given by $\phi(n) = x^n$ is a homomorphism. It follows from part 2 of Theorem 10.2 that there are no others.
63. Let ϕ be a homomorphism from S_3 to Z_n . Since $|\phi(S_3)|$ must divide 6 we have that $|\phi(S_3)| = 1, 2, 3$, or 6. In the first case ϕ maps every element to 0. If $|\phi(S_3)| = 2$, then n is even and ϕ maps the even permutations to 0, and the odd permutations to $n/2$. The case that $|\phi(S_3)| = 3$ cannot occur because it implies that $\text{Ker } \phi$ is a normal subgroup of order 2 whereas S_3 has no normal subgroup of order 2. The case that $|\phi(S_3)| = 6$ cannot occur because it implies that ϕ is an isomorphism from a non-Abelian group to an Abelian group.

64. Suppose that ϕ is a homomorphism and $m = nq + r$ where $1 \leq r < n$. Then $0 = \phi(0) = \phi(nq + r) = \phi(nq) + \phi(r) = r \neq 0$. This contradiction shows that for ϕ to be a homomorphism it is necessary that n divides m . If n divides m , then ϕ is well-defined and is operation preserving from Exercise 9 of Chapter 0.
65. $\phi(zw) = z^2w^2 = \phi(z)\phi(w)$. $\text{Ker } \phi = \{1, -1\}$ and, because ϕ is onto $\mathbf{C}^*$, we have by Theorem 10.3, $\mathbf{C}^*/\{1, -1\}$ is isomorphic to $\mathbf{C}^*$. If $\mathbf{C}^*$ is replaced by $\mathbf{R}^*$ we have that ϕ is onto $\mathbf{R}^+$ and by Theorem 10.3, $\mathbf{R}^*/\{1, -1\}$ is isomorphic to $\mathbf{R}^+$.
66. p^2 . To verify this note that for any homomorphism ϕ from $Z_p \oplus Z_p$ into Z_p we have $\phi(a, b) = a\phi(1, 0) + b\phi(0, 1)$. Thus we need only count the number of choices for $\phi(1, 0)$ and $\phi(0, 1)$. Since p is prime we may let $\phi(1, 0)$ be any element of Z_p . The same is true for $\phi(0, 1)$.

CHAPTER 11

Fundamental Theorem of Finite Abelian Groups

1. $n = 4$
 $Z_4, Z_2 \oplus Z_2$
2. $n = 8$; $Z_8, Z_4 \oplus Z_2, Z_2 \oplus Z_2 \oplus Z_2$
3. $n = 36$
 $Z_9 \oplus Z_4, Z_3 \oplus Z_3 \oplus Z_4, Z_9 \oplus Z_2 \oplus Z_2, Z_3 \oplus Z_3 \oplus Z_2 \oplus Z_2$
4. order 2: 1, 3, 3, 7; order 4: 2, 4, 12, 8
5. The only Abelian groups of order 45 are Z_{45} and $Z_3 \oplus Z_3 \oplus Z_5$. In the first group, $|3| = 15$; in the second one, $|(1, 1, 1)| = 15$. $Z_3 \oplus Z_3 \oplus Z_5$ does not have an element of order 9.
6. $Z_{27} \oplus Z_4; Z_{27} \oplus Z_2 \oplus Z_2$
7. In order to have exactly four subgroups of order 3, the group must have exactly 8 elements of order 3. When counting elements of order 3 we may ignore the components of the direct product that represent the subgroup of order 4 since their contribution is only the identity. Thus, we examine Abelian groups of order 27 to see which have exactly 8 elements of order 3. By Theorem 4.4, Z_{27} has exactly 2 elements of order 3; $Z_9 \oplus Z_3$ has exactly 8 elements of order 3 since for $|(a, b)| = 3$ we can choose $|a| = 1$ or 3 and $|b| = 1$ or 3, but not both $|a|$ and $|b|$ of order 1; In $Z_3 \oplus Z_3 \oplus Z_3$ every element except the identity has order 3. So, the Abelian groups of order 108 that have exactly four subgroups of order 3 are $Z_9 \oplus Z_3 \oplus Z_4$ and $Z_9 \oplus Z_3 \oplus Z_2 \oplus Z_2$. The subgroups of $Z_9 \oplus Z_3 \oplus Z_4$ of order 3 are $\langle(3, 0, 0)\rangle, \langle(0, 1, 0)\rangle, \langle(3, 1, 0)\rangle$ and $\langle(3, 2, 0)\rangle$. The subgroups of $Z_9 \oplus Z_3 \oplus Z_2 \oplus Z_2$ of order 3 are $\langle(3, 0, 0, 0)\rangle, \langle(0, 1, 0, 0)\rangle, \langle(3, 1, 0, 0)\rangle$ and $\langle(3, 2, 0, 0)\rangle$.
8. $Z_3 \oplus Z_3 \oplus Z_3 \oplus Z_4; Z_3 \oplus Z_3 \oplus Z_3 \oplus Z_2 \oplus Z_2$
9. Elements of order 2 are determined by the factors in the direct product that have order a power of 2. So, we need only look at $Z_8, Z_4 \oplus Z_2$ and $Z_2 \oplus Z_2 \oplus Z_2$. By Theorem 4.4, Z_8 has exactly one element of order 2; $Z_4 \oplus Z_2$ has exactly three elements of order 2; $Z_2 \oplus Z_2 \oplus Z_2$ has exactly 7 elements of order 2. So, $G \approx Z_4 \oplus Z_2 \oplus Z_3 \oplus Z_5$.
10. $Z_8 \oplus Z_9 \oplus Z_5; Z_4 \oplus Z_2 \oplus Z_9 \oplus Z_5; Z_2 \oplus Z_2 \oplus Z_2 \oplus Z_9 \oplus Z_5; Z_8 \oplus Z_3 \oplus Z_3 \oplus Z_5; Z_4 \oplus Z_2 \oplus Z_3 \oplus Z_3 \oplus Z_5; Z_2 \oplus Z_2 \oplus Z_2 \oplus Z_3 \oplus Z_3 \oplus Z_5$.

11. By the Fundamental Theorem, any finite Abelian group G is isomorphic to some direct product of cyclic groups of prime-power order. Now go across the direct product and, for each distinct prime you have, pick off the largest factor of that prime-power. Next, combine all of these into one factor (you can do this, since their orders are relatively prime). Let us call the order of this new factor n_1 . Now repeat this process with the remaining original factors and call the order of the resulting factor n_2 . Then n_2 divides n_1 , since each prime-power divisor of n_2 is also a prime-power divisor of n_1 . Continue in this fashion. Example: If

$$G \approx Z_{27} \oplus Z_3 \oplus Z_{125} \oplus Z_{25} \oplus Z_4 \oplus Z_2 \oplus Z_2,$$

then

$$G \approx Z_{27 \cdot 125 \cdot 4} \oplus Z_{3 \cdot 25 \cdot 2} \oplus Z_2.$$

Now note that 2 divides $3 \cdot 25 \cdot 2$ and $3 \cdot 25 \cdot 2$ divides $27 \cdot 125 \cdot 4$.

12. By the corollary to the Fundamental Theorem of Finite Abelian Groups the given group has a subgroup of order 10. But this group must be isomorphic to $Z_2 \oplus Z_5 \approx Z_{10}$.
13. $Z_2 \oplus Z_2$
14. If G is an Abelian group of order n and m is a divisor of n , then G has a cyclic subgroup of order m if m is squarefree (i.e., each prime factor of m occurs to the 1st power only).
15. **a.** 1 **b.** 1 **c.** 1 **d.** 1 **e.** 1 **f.** There is a unique Abelian group of order n if and only if n is not divisible by the square of any prime.
16. **a.** same **b.** same **c.** same **d.** same
e. twice as many of order m compared with the number of order n
17. This is equivalent to asking how many Abelian groups of order 16 have no element of order 8. From the Fundamental Theorem of Finite Abelian Groups the only choices are $Z_4 \oplus Z_4$, $Z_4 \oplus Z_2 \oplus Z_2$, and $Z_2 \oplus Z_2 \oplus Z_2 \oplus Z_2$.
18. 5^n
19. The symmetry group is $\{R_0, R_{180}, H, V\}$. Since this group is Abelian and has no element of order 4, it is isomorphic to $Z_2 \oplus Z_2$.
20. Consider every possible isomorphism class one by one and show each has the desired subgroup. For instance, in $Z_2 \oplus Z_2 \oplus Z_2 \oplus Z_{27} \oplus Z_5$ the subgroup is
- $$\{(a, b, 0, c, d) \mid a, b \in Z_2, c \in \langle 3 \rangle, d \in Z_5\}$$
21. Because the group is Abelian and has order 9, the only possibilities are Z_9 and $Z_3 \oplus Z_3$. Since Z_9 has exactly 2 elements of order 3 and 9, 16, and 22 have order 3, the group must be isomorphic to $Z_3 \oplus Z_3$.

22. Because of the Fundamental Theorem and Corollary 1 of Theorem 8.2, we may assume $|G|$ is a prime-power. Let x be an element of G of maximum order. Then for any y in G we have $|y|$ divides $|\langle x \rangle|$. Since $\langle x \rangle$ has a unique subgroup for each divisor of $|\langle x \rangle|$ it follows that $\langle y \rangle \subseteq \langle x \rangle$.
23. By the Corollary of Theorem 8.2, n must be square-free (no prime factor of n occurs more than once).
24. $n = p_1^2 p_2^2$ or $p_1^2 p_2^2 p_3 p_4 \cdots p_k$ where $k \geq 3$ and $p_1, p_2, \dots, p_k$ are distinct primes.
25. Among the first 11 elements in the table, there are 9 elements of order 4. None of the other isomorphism classes has this many.
26. $Z_4 \oplus Z_2$; One internal direct product is $\langle 7 \rangle \times \langle 17 \rangle$.
27. First observe that G is Abelian and has order 16. Now we check the orders of the elements. Since the group has 8 elements of order 4 and 7 of order 2 it is isomorphic to $Z_4 \oplus Z_2 \oplus Z_2$. One internal direct product is $\langle 7 \rangle \times \langle 101 \rangle \times \langle 199 \rangle$.
28. $Z_2 \oplus Z_2 \oplus Z_3$; One internal direct product is $\langle 19 \rangle \times \langle 26 \rangle \times \langle 31 \rangle$.
29. Since Z_9 has exactly 2 elements of order 3 once we choose 3 nonidentity elements we will either have at least one element of order 9 or 3 elements of order 3. In either case we have determined the group. The Abelian groups of order 18 are $Z_9 \oplus Z_2 \approx Z_{18}$ and $Z_3 \oplus Z_3 \oplus Z_2$. By Theorem 4.4, Z_{18} group has 6 elements of order 18, 6 elements of order 9, 2 of order 6, 2 of order 3, 1 of order 2, and 1 of order 1. $Z_3 \oplus Z_3 \oplus Z_2$ has 8 elements of order 3, 8 of order 6, 1 of order 2, and 1 of order 1. The worst case scenario is that at the end of 5 choices we have selected 2 of order 6, 2 of order 3, and 1 of order 2. In this case we still have not determined which group we have. But the sixth element we select will give us either an element of order 18 or 9, in which case we know the group Z_{18} or a third element of order 6 or 3, in which case we know the group is $Z_3 \oplus Z_3 \oplus Z_2$.
30. The element of order 8 rules out all but Z_{16} and $Z_8 \oplus Z_2$ and two elements of order 2 precludes Z_{16} .
31. If $a^2 \neq b^2$, then $a \neq b$ and $a \neq b^3$. It follows that $\langle a \rangle \cap \langle b \rangle = \{e\}$. Then $G = \langle a \rangle \times \langle b \rangle \approx Z_4 \oplus Z_4$.
32. If G is an Abelian group of order 2^n observe that the elements of G order 2 together with the identity form a subgroup H . Then if $|H| = 2^m$, then By Theorem 11.1, H is isomorphic to $Z_2 \times Z_2 \times \cdots \times Z_2$ (m copies). Thus the number of elements of order 2 in G is $2^m - 1$.
33. By Theorem 11.1, we can write the group in the form $Z_{p_1^{n_1}} \oplus Z_{p_2^{n_2}} \oplus \cdots \oplus Z_{p_k^{n_k}}$ where each p_i is an odd prime. By Theorem 8.1

the order of any element $(a_1, a_2, \dots, a_k) = \text{lcm}(|a_1|, |a_2|, \dots, |a_k|)$. And from Theorem 4.3 we know that $|a_i|$ divides $p_i^{n_i}$, which is odd.

34. $Z_2 \oplus Z_2 \oplus \dots \oplus Z_2$ (n terms)
35. By Theorem 7.2 we have,
 $|\langle a \rangle K| = |a||K|/|\langle a \rangle \cap K| = |a||K| = |\bar{a}||\bar{K}|p = |\bar{G}|p = |G|$.
36. If $|G| = p^n$, use the Fundamental Theorem and Theorem 9.6. If every element has order a power of p use the corollary to the Fundamental Theorem.
37. By the Fundamental Theorem of Finite Abelian Groups, it suffices to show that every group of the form $Z_{p_1^{n_1}} \oplus Z_{p_2^{n_2}} \oplus \dots \oplus Z_{p_k^{n_k}}$ is a subgroup of a U -group. Consider first a group of the form $Z_{p_1^{n_1}} \oplus Z_{p_2^{n_2}}$ (p_1 and p_2 need not be distinct). By Dirichlet's Theorem, for some s and t there are distinct primes q and r such that $q = tp_1^{n_1} + 1$ and $r = sp_2^{n_2} + 1$. Then $U(qr) = U(q) \oplus U(r) \approx Z_{tp_1^{n_1}} \oplus Z_{sp_2^{n_2}}$, and this latter group contains a subgroup isomorphic to $Z_{p_1^{n_1}} \oplus Z_{p_2^{n_2}}$. The general case follows in the same way.
38. Observe that
 $\text{Aut}(Z_2 \oplus Z_3 \oplus Z_5) \approx \text{Aut}(Z_{30}) \approx U(30) \approx U(2) \oplus U(3) \oplus U(5) \approx Z_2 \oplus Z_4$.
39. It follows from Exercise 4 of Chapter 8 and Theorem 9.6 that if D_4 could be written in the form $\langle a \rangle \times K$ where $|a| = 4$ it would be Abelian.